

Radiation dose from contemporary cardiothoracic multidetector CT protocols with an anthropomorphic female phantom: implications for cancer induction.



PURPOSE: To measure prospectively and directly both organ dose and effective dose (ED) for adult cardiac and pulmonary computed tomographic (CT) angiography by using current clinical protocols for 64-detector CT in an anthropomorphic female phantom and to estimate lifetime attributable risk of breast and lung cancer incidence on the basis of measured ED and organ dose. MATERIALS AND METHODS: Cardiac and pulmonary 64-detector CT angiography was performed by using current clinical protocols to evaluate the pulmonary veins (electrocardiographically [ECG] gated, 64 sections at 0.625-mm collimation, 120 kVp, 300 mA, 0.35-second tube rotation), native coronary arteries (ECG gated; 64 sections at 0.625 mm; 120 kVp; maximum current, 500-750 mA; minimum, 100-350 mA; 0.35-second tube rotation) and pulmonary embolus (64 sections at 1.25 mm, 140 kVp, 645 mA, 0.5-second tube rotation). Absorbed organ doses were measured by using an anthropomorphic female phantom and metal oxide semiconductor field effect transistor detectors. ED was calculated from measured organ doses and the dose-length product. RESULTS: ED for current adult cardiac and pulmonary 64-detector CT angiography protocols were 12.4-31.8 mSv. Overall, skin, breast, and esophagus and heart had the highest recorded absorbed organ doses. Relative risk for breast cancer incidence for girls and women was 1.004-1.042 for a single examination. Relative risk for lung cancer incidence for men and women was 1.005-1.076 from a single examination. CONCLUSION: EDs and organ doses from 64-detector CT are higher than those previously reported for adult cardiac and pulmonary CT angiography protocols. Risk for breast and lung cancer induction from these studies is greatest for the younger patient population. (c) RSNA, 2007.